Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 The Born rule is equivalent to the von Neumann entropy

Files.

- File: `briefs/BornRuleEntropy/BornRuleEntropy.lean`
- Library: `BornRuleEntropy`
- Imports: `ForPhyslib.QuantumMechanics.VonNeumannEntropyMixture`
- New `ForPhyslib` material:
  `ForPhyslib/QuantumMechanics/VonNeumannEntropyMixture.lean`: a mixture $\sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$ of orthonormal pure states, in any dimension, is a density operator (`isDensityOperator_tsum_smul_proj`) with entropy $\sum_i \eta(p_i)$, $\eta(x) = -x \log x$ (`vonNeumannEntropy_tsum_smul_proj`); the entropy of an equal mixture of two pure states is $h((1 + |\left\langle\psi|\phi\right\rangle|)/2)$, h the binary entropy (`vonNeumannEntropy_mix_proj`).

Setting. E is a complex Hilbert space of any dimension. A pure state is $|\psi\rangle\langle\psi|$ (`proj` ψ) for a unit vector ψ, and an ensemble is a density operator (`IsDensityOperator`: positive, trace class, trace one). The entropy S is a function from operators to $[0,\infty]$, and the von Neumann entropy `vonNeumannEntropy` is $\operatorname{tr}\eta(\rho) \in [0,\infty]$ with the natural logarithm, so the brief's "$S = 1$" (one bit) reads $S = \log 2$. The conditional probability is a function p of two operators; $p(|\psi\rangle\langle\psi|, \rho)$ is the probability of finding ψ having prepared ρ. The conditions are:

- `EnsemblesAndStates` p (EQST): $p(|\psi\rangle\langle\psi|, \rho) \in [0,1]$ for density operators ρ; $p(|\psi\rangle\langle\psi|, t|\phi_1\rangle\langle\phi_1| + (1-t)|\phi_2\rangle\langle\phi_2|) = t\,p(|\psi\rangle\langle\psi|, |\phi_1\rangle\langle\phi_1|) + (1-t)\,p(|\psi\rangle\langle\psi|, |\phi_2\rangle\langle\phi_2|)$ for $t \in [0,1]$; and $p(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = 1$ iff $|\psi\rangle\langle\psi| = |\phi\rangle\langle\phi|$;
- `BornRule` p (BR-BORN): $p(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2$;
- `IsVonNeumannEntropy` S (BR-ENT): $S(\rho) =$ `vonNeumannEntropy` ρ for density operators ρ;
- `MutualExclusivity` S p ([B]MUTEX): for every countable family (ψ_i) of unit vectors, $p(|\psi_i\rangle\langle\psi_i|, |\psi_j\rangle\langle\psi_j|) = 0$ for all $i \neq j$ iff $S(\sum_i w_i |\psi_i\rangle\langle\psi_i|) = \sum_i \eta(w_i)$ for all weights $w_i \geq 0$ with $\sum_i w_i = 1$ (the sum of operators converging in operator norm).

Results.

Brief	Lean
Proposition 3, eq. (6), $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = h((1+\sqrt{\lambda})/2)$, $\lambda = \lvert\langle\psi\vert\phi\rangle\rvert^2$	`vonNeumannEntropy_mix_proj_eq` (from `vonNeumannEntropy_mix_proj`)
Proposition 3, $p(\phi\vert\psi) = \lambda$ iff $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = h((1+\sqrt{\lambda})/2)$	`norm_inner_sq_eq_iff`, for $\lambda \in [0,1]$, with p the Born rule
Fig. 2, the entropy is strictly decreasing in the Born probability	`strictAntiOn_entropyOfBorn`
Corollary 7, $\langle\phi\vert\psi\rangle = 0$ iff $S(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi) = 1$	`inner_eq_zero_iff`, with $\log 2$ for 1
EQST, BR-BORN, BR-ENT, [B]MUTEX	`EnsemblesAndStates`, `BornRule`, `IsVonNeumannEntropy`, `MutualExclusivity`
Theorem 9, BR-BORN and BR-ENT imply [B]MUTEX	`MutualExclusivity.of_bornRule`
Theorem 9, BR-BORN and [B]MUTEX imply BR-ENT	`IsVonNeumannEntropy.of_bornRule`
Theorem 9, BR-ENT and [B]MUTEX imply BR-BORN	`BornRule.of_isVonNeumannEntropy`, over EQST
Theorem 9	`two_conditions_imply_third`
(used for the last direction) orthogonal states are mutually exclusive	`apply_eq_zero_of_inner_eq_zero`
(used for the last direction) the Born rule on the qubit spanned by ψ and $\chi \perp \psi$	`apply_planeVec_eq`: $p(\psi\vert r\psi + s\bar{\zeta}\chi) = r^2$; from `apply_planeVec` and `apply_halfVec`

Differences from the brief.

- E is a Hilbert space of any dimension; separability is not used. [B]MUTEX quantifies over countable index types (in the universe of E). For BR-BORN + [B]MUTEX $\Rightarrow$ BR-ENT the

eigenbasis of ρ is restricted to the eigenvectors with nonzero eigenvalue, a countable family since the eigenvalues are summable.

- Entropies take values in $[0, \infty]$, not in $\mathbb{R}$: the von Neumann entropy and the Shannon entropy of a countable family of weights can be infinite (see "Problems found").
- In EQST, affinity is assumed only for mixtures of two pure states, which is weaker than the brief's countable mixtures. EQST is used only for BR-ENT + [B]MUTEX $\Rightarrow$ BR-BORN, and there only through $p(\psi|\psi) = 1$, $p \geq 0$ on pure states, and affinity for two pure states. The other two directions do not use EQST. Corollary 8 is not formalized.
- The first argument of p is the projection $|\psi\rangle\langle\psi|$, so p depends on the ray of ψ only.
- In BR-ENT + [B]MUTEX $\Rightarrow$ BR-BORN, [B]MUTEX is used only for pairs of orthogonal states, and only in the direction "Shannon entropy $\Rightarrow$ mutually exclusive". In BR-BORN + [B]MUTEX $\Rightarrow$ BR-ENT only the other direction is used.
- The Bloch-ball argument for BR-BORN (Fig. 3) is replaced by an algebraic one in the qubit spanned by ψ and a unit $\chi \perp \psi$. With $f = p(\psi|\cdot)$, $f(\psi) = 1$ and $f(\chi) = 0$, the convex identities $|u\rangle\langle u| + s(r-s)\,|\chi\rangle\langle\chi| = r(r-s)\,|\psi\rangle\langle\psi| + 2rs\,|e_\zeta\rangle\langle e_\zeta|$ (for $r \geq s$, and symmetrically for $r \leq s$), with $u = r\psi + s\bar{\zeta}\chi$ and $e_\zeta = (\psi + \bar{\zeta}\chi)/\sqrt{2}$, give $f(u) = r^2 + 2rs(f(e_\zeta) - \frac{1}{2})$, and $|e_\zeta\rangle\langle e_\zeta| + |e_{-\zeta}\rangle\langle e_{-\zeta}| = |\psi\rangle\langle\psi| + |\chi\rangle\langle\chi|$ gives $f(e_\zeta) + f(e_{-\zeta}) = 1$. If $d = f(e_\zeta) - \frac{1}{2} < 0$, the state with $r/s = -d$ has $f(u) = -d^2/(1+d^2) < 0$, so $f(e_\zeta) = \frac{1}{2}$ and $f(u) = r^2 = |\langle\psi|u\rangle|^2$.

Problems found in the brief. None that affect Theorem 9; some statements need care.

- EQST takes $S : \mathcal{E}(\mathcal{H}) \to \mathbb{R}$, but in infinite dimension the von Neumann entropy is $+\infty$ for some density operators, and $-\sum_i p_i \log p_i$ can be $+\infty$ for countably many weights. With a real-valued S, BR-ENT and [B]MUTEX cannot hold in infinite dimension, and Theorem 9 is true there only vacuously. The formalization uses $[0, \infty]$.
- The units are not fixed: eq. (2) writes log, while Corollary 7 and Fig. 2 use entropy 1 for two orthogonal states, which needs $\log_2$.
- BR-ENT calls $-\operatorname{tr}(\rho \log \rho)$ "the Shannon entropy".
- The proof of BR-ENT + [B]MUTEX $\Rightarrow$ BR-BORN refers to "the proof of 1" and "the same trigonometric arguments in the proof of 1"; there is no item 1, and Proposition 3 and Corollary 7 are meant. It also says "p must take its extreme values on ψ and ϕ", where ψ and $\psi^\perp$ are meant.
- In the same proof, "Given that the range of p is $[0, 1]$, p must take its extreme values on ψ and $\psi^\perp$. Therefore, ψ and $\psi^\perp$ must be opposite points on the ball" does not say why the affine function $p(\psi|\cdot)$ has its gradient along the axis through ψ and $\psi^\perp$; that two orthogonal states are antipodal already holds in the Bloch ball of EQST. The gradient is fixed because $p \geq 0$ near $\psi^\perp$, where $p = 0$; the formalization makes this explicit ("Differences", last item).